ENTERPRISE COUNTER-UAS (C-UAS) PLATFORM

Drone Threat Command & Detection System

Full Technical Architecture, Engineering Deep-Dive & Interview Master Guide

1. EXECUTIVE SUMMARY & THREAT LANDSCAPE

Low-altitude micro and commercial drones (UAS) present acute security vulnerabilities for critical infrastructure - including commercial airports, military defense perimeters, government buildings, energy grids, and public stadiums. Small radar cross-sections (RCS) and agile flight allow unauthorized drones to bypass traditional radar surveillance.

The Drone Threat Command & Detection System is a high-performance counter-unmanned aircraft system (C-UAS) integrating deep learning computer vision, centroid multi-object tracking, multi-sensor data fusion, dynamic geofencing, threat risk scoring, and tactical electronic countermeasures into a centralized operator dashboard.

2. HIGH-LEVEL SYSTEM ARCHITECTURE & DATA FLOW

The system operates across a modular four-tier architecture designed for low latency (<40ms) and multi-sensor resilience:

Layer / Subsystem	Primary Technologies	Core Operational Responsibility
1. Perception Layer	OpenCV, YOLOv8, SDR / RF, Radar Scope	Raw video ingestion, RF spectrum scanning (2.4/5.8GHz), Doppler radar simulation.
2. Tracking & Fusion	Centroid Tracker, NumPy, Trajectory Cache	Target association, velocity estimation (px/frame), and multi-sensor telemetry fusion.
3. Geofence & Risk	Coordinate Mapping, Risk Scoring Engine	Polygon boundary checks, inbound heading calculation, and 0-100 threat categorizati
4. Defense & C2	FastAPI, React 18, Web Audio, MJPEG	Countermeasure dispatch (RF Jamming / GNSS / ATC), live streaming, and reporting

KEY ARCHITECTURAL HIGHLIGHT: ZERO-LATENCY MJPEG STREAMING

- Asynchronous FastAPI multipart replacement streaming (multipart/x-mixed-replace; boundary=frame).
- Yields JPEG byte streams natively rendered by HTML5 <img> tags with zero client-side buffering latency.
- Seamlessly switches between live camera hardware and high-fidelity simulated tactical drone footage.